



Korognai SMO

Reference Implementation — Full Documentation

A single-file, browser-based demonstration of a Service Management & Orchestration (SMO) platform for O-RAN networks — every tab, every term, every scenario explained.

Updated July 2026 · Synthetic demo, no live network or vendor affiliation

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1. Purpose & How to Use This Document

This document explains, in full detail, every tab, metric, term, and simulated scenario inside the Korognai SMO reference demo. It is written for two audiences at once: someone with deep telecom background who wants to verify the technical grounding, and someone with no telecom background who wants to understand what they're looking at and why it matters.

What this demo is: a single, self-contained HTML file that runs entirely in a web browser, with no backend server and no live network behind it. Every number, alarm, and event on screen is generated live, in-browser, by JavaScript — it is a working simulation of how a real SMO platform behaves, not a recording or a mockup of static screenshots.

What this demo is not: it is not connected to any real radio network, it does not use any vendor's proprietary software or confidential material, and none of the named companies, products, or trademarks of any commercial SMO vendor appear in it. The demo uses only generic, industry-standard terminology drawn from public O-RAN Alliance, 3GPP, TM Forum, NIST, and CIS material.

The rest of this document walks through the demo in the same order the tabs appear, left to right, followed by two tabs added since the previous edition of this document — Documentation and 50 Design Questions — and closes with the Design Principles and Appendix. Section 2 defines every acronym and concept used anywhere in the demo, so later sections can reference them without re-explaining each time.

What's new since the previous edition: six working rApps instead of four (two previously-decorative rApps, Automatic Neighbor Relation and PCI Reuse Optimization, now have genuine trigger logic and their own dedicated scenario buttons); live, automatic rApp-vs-rApp conflict detection wired into the Conflict Log's resolution strategies; a fully interactive, select-an-object model for Cloud & NF Lifecycle with infrastructure that can spontaneously degrade and self-resolve; ten live KPIs instead of four, each with a hover-revealed trend and min/max; a 100-entry, scrollable Fault Management, Configuration Management, and Trace Sessions view; and expanded in-tab explanations of exactly where each tab's inputs come from, rather than treating automation as a black box.

2. Core Concepts & Glossary

2.1 The physical radio network, in plain terms

A mobile network's radio side used to be one box at each cell tower. Modern networks split that single box into three functional pieces so they can be built more flexibly and partly run in the cloud:

TERM	FULL NAME	PLAIN-ENGLISH MEANING
RU	Radio Unit	The physical hardware at the antenna. Converts digital signals into actual radio waves and back. Stays at the tower.
DU	Distributed Unit	Handles time-critical, millisecond-scale processing. Usually located physically close to the tower because timing is so strict.
CU	Centralized Unit	Handles less time-critical processing and can be centralized to serve many towers at once. Split into CU-CP (control) and CU-UP (user data).
Core	Core Network	The wired part of the network the radio side connects into — routes traffic to the internet or other networks.

Transport segments — the links between these pieces have their own names, because their timing and bandwidth requirements are all different:

SEGMENT	CONNECTS	TIMING SENSITIVITY
Fronthaul	RU ↔ DU	Extremely high — microseconds
Midhaul	DU ↔ CU	High — single-digit milliseconds
Backhaul	CU ↔ Core	Moderate

2.2 The management & intelligence layers

TERM	FULL NAME	PLAIN-ENGLISH MEANING
SMO	Service Management & Orchestration	The top-level management "brain" that oversees the whole radio network from above — monitoring, configuring, securing, and automating it.
Non-RT RIC	Non-Real-Time RAN Intelligent Controller	The slower (greater than 1 second, often minutes+) intelligence layer that lives inside the SMO. Hosts rApps. Handles longer-term optimization and AI/ML model training.
Near-RT RIC	Near-Real-Time RAN Intelligent Controller	A separate platform, physically closer to the radio, making real-time decisions on a 10-millisecond-to-1-second timescale — fast enough that a human, or even the Non-RT RIC, couldn't react in time. Hosts xApps.
rApp	(RIC app)	An automated behavior/application hosted on the Non-RT RIC — e.g. energy saving, mobility tuning. Slower timescale, more sophisticated analysis.
xApp	(RIC app)	An automated real-time behavior hosted on the Near-RT RIC — e.g. dynamic spectrum sharing. Fast timescale, simpler/quicker decisions.

2.3 Interfaces — how the layers talk to each other

O-RAN Alliance defines a set of open, standardized interfaces so that equipment from different vendors can interoperate. This is the single most important reference table for understanding how any part of the demo connects to any other.

INTERFACE	CONNECTS	PURPOSE
O-FH	RU ↔ DU	Standardized Open Fronthaul, based on a specific processing "cut point" called functional split 7.2x — enables multi-vendor RU/DU interoperability.
F1	DU ↔ CU	Midhaul interface, split into F1-C (control) and F1-U (user plane).
E1	CU-CP ↔ CU-UP	Separates control-plane and user-plane processing within the CU.
E2	Near-RT RIC ↔ RU/DU/CU	Carries real-time RAN data up, and real-time control actions back down. The xApp control channel.
A1	Non-RT RIC (SMO) ↔ Near-RT RIC	Carries AI/ML-informed policy guidance and enrichment data down from the slower layer to the faster one.
O1	SMO ↔ all managed RAN elements	FCAPS — fault, configuration, accounting, performance, and security management.
O2	SMO ↔ O-Cloud infrastructure	Cloud infrastructure and workload orchestration/lifecycle management.
R1	rApps ↔ Non-RT RIC	How an rApp registers itself, declares what data it needs (via SME/DME), and receives that data — still being

2.4 FCAPS — the five pillars of network monitoring

LETTER	STANDS FOR	PLAIN-ENGLISH MEANING
F	Fault Management	Tracking things that have gone wrong (alarms), so they can be acknowledged, cancelled once resolved, and fixed.
C	Configuration Management	A record of every setting on the network and every change made to it.
A	Accounting Management	Tracking usage and resource consumption (often for billing or capacity planning).
P	Performance Management	Tracking how well the network is actually running — speed, reliability, capacity.
S	Security Management	Access control, audit trails, and protecting the network from misuse.

2.5 Radio, performance & 5G traffic-type terms used throughout the demo

TERM	PLAIN-ENGLISH MEANING
PRB	Physical Resource Block — the basic unit of radio capacity a cell tower has to allocate to users. "PRB utilization" is how much of that capacity is currently in use — like asking what percentage of a highway's lanes are occupied.
Handover (HO)	What happens when a phone moves and switches from one cell tower's coverage to another's, automatically, without dropping the call. A "handover failure" is often experienced as a dropped call or a brief lag.
RRC	Radio Resource Control — the signaling protocol a device uses to establish its radio-level connection to a cell in the first place, before any handover ever happens.
RSRP	Reference Signal Received Power — a measurement of how strong a phone's signal from a given tower actually is.
UE	User Equipment — the device connecting to the network (phone, sensor, industrial modem, etc.).
eMBB	Enhanced Mobile Broadband — the 5G use case optimized for raw speed and capacity, as opposed to latency or device density.
URLLC	Ultra-Reliable Low-Latency Communication — the 5G use case optimized for guaranteed, near-instant delivery, even at the cost of raw throughput.
mMTC	Massive Machine-Type Communication — the third 5G use-case pillar, for huge numbers of low-power sensors/devices at once. Named here for completeness alongside eMBB/URLLC.
GBR / PDB / PER	Guaranteed Bit Rate (reserved capacity for a flow); Packet Delay Budget (max acceptable one-way delay); Packet Error Rate (fraction of packets lost or corrupted — URLLC targets a near-zero rate).
CA	Carrier Aggregation — combining multiple frequency channels into one connection to boost peak throughput.
DL / UL	Downlink (network → device) and Uplink (device → network) — the two directions of radio traffic.
Drop Call Rate	The percentage of calls that get disconnected unexpectedly mid-call.
Call Setup Success Rate	The percentage of attempts to start a call (or data session) that succeed on the first try.
Network Availability	The percentage of time the network is actually up and reachable. Capped at 100% in the demo, because it is a percentage of uptime and no system can be "available" more than all of the time.

2.6 Automation concepts

TERM	PLAIN-ENGLISH MEANING
Closed loop	A fully automated cycle: observe a condition → decide what to do → act → verify the result — with no human step in between. Contrast with "open loop," where the system only recommends an action for a human to approve.
SON	Self-Organizing Networks — the long-established industry term for exactly this kind of automated, closed-loop network optimization. rApps are essentially SON formalized under open, standardized interfaces.
Conflict management	What happens when two automated behaviors try to change the same thing at the same time in different directions. See Section 11 for the three resolution strategies used in this demo, and how genuine conflicts are now detected live rather than only demonstrated on request.
Intent-based management	Telling the network what outcome you want (e.g. "guarantee low latency for this customer") rather than how to achieve it — the system translates that intent into the actual technical settings.
Agentic AI	AI that can reason, plan, and take multi-step action toward a goal, rather than just following a fixed rule. The demo's "Copilot narration" feature is a deliberately modest, honest example: it explains decisions in plain language rather than claiming full autonomous reasoning.

2.7 Network slicing terms

TERM	PLAIN-ENGLISH MEANING
Network slice	A virtual, separate network carved out of one shared physical network, with its own guarantees — e.g. a guaranteed-speed lane for one enterprise customer, running alongside everyone else's regular traffic on the same towers.
NSSI	Network Slice Subnet Instance — the specific piece of a slice that lives in the radio access network, as opposed to the core network or transport.
S-NSSAI	The identifier a network uses internally to tell which slice a piece of traffic belongs to.
5QI	A standardized code that tells the network what quality-of-service treatment (delay tolerance, priority, reliability) a particular type of traffic needs.
SLA	Service Level Agreement — a measurable, contractual promise about network quality (e.g. "99.9% uptime," "under 10ms latency").

2.8 Cloud & infrastructure terms

TERM	PLAIN-ENGLISH MEANING
O-Cloud	The generic O-RAN term for the cloud computing infrastructure that hosts virtualized network functions, as opposed to dedicated, single-purpose hardware.
CNF	Cloud-native Network Function — network software packaged to run in containers (like Kubernetes), the modern alternative to dedicated hardware boxes.
vDU / vCU-CP / vCU-UP	Virtualized versions of the DU and CU, running as software on standard servers instead of dedicated hardware.
NF Lifecycle (LCM)	The full set of operations a network function goes through: instantiate (stand up new), heal (auto-recover from a fault), scale (add/remove capacity), upgrade (new software, no downtime), and terminate (clean shutdown). See Section 13 for how this is now driven by selecting a real object rather than triggering an abstract action.
O2-DMS	O2 Deployment Management Service — the specific interface the SMO uses to tell the cloud infrastructure what software to actually run.
PnP	Plug-and-Play — new hardware announces itself to the management system and gets configured automatically, with no manual setup step.
DC	Data Center — the physical or regional facility hosting cloud-native network functions (e.g. vCU, vDU).

2.9 Security & trust terms

TERM	PLAIN-ENGLISH MEANING
Zero-trust (NIST SP 800-207)	A security model where nothing is trusted automatically — every request is checked every time, even from inside the network perimeter.
CIS Benchmarks	An independent, published checklist of secure configuration settings for common software — used to verify a system is actually hardened, not just assumed to be.
SBOM	Software Bill of Materials — a complete, itemized list of every component inside a piece of software, similar to an ingredients label on food.
Supply-chain integrity	Proof that the software running today is exactly what was built and approved, with nothing tampered with along the way — verified through digital signatures and checksums against a build-time attestation.

2.10 Business & commercial terms

TERM	PLAIN-ENGLISH MEANING
TCO	Total Cost of Ownership — the full lifecycle cost of a solution (purchase, energy, labor, integration, maintenance), not just the sticker price.
OPEX	Operating Expenses — the ongoing cost of running something, as distinct from one-time capital purchase cost.
OSS / BSS	Operations Support System / Business Support System — the operator's own systems (network ops and billing/customer-facing respectively) that sit north of the SMO and originate many of the real-world orders (slice orders, service terminations) this demo models.
SME / DME	Service Management & Exposure / Data Management & Exposure — the R1 services an rApp uses to discover, request, and consume data and management functions from the Non-RT RIC.
MTTR	Mean Time to Resolve — how long, on average, between a problem starting and it actually being fixed, whether by automation or by a person.

3. Overview Tab

This is the landing page and is deliberately written for a non-technical reader — the goal is that a VP or an operations executive can understand what the platform does within about thirty seconds of opening it, before any of the deeper tabs are explored.

3.1 "In plain terms" callout

A single paragraph at the very top explains the whole platform as "the control room that runs a modern mobile network" — spotting problems before customers notice, fixing many automatically, cutting energy costs, and giving operations staff one screen instead of a dozen disconnected tools. This framing deliberately avoids all acronyms.

3.2 Executive KPI row

KPI	WHAT IT MEASURES	WHERE THE NUMBER COMES FROM
Issues Resolved Automatically	How many times an rApp actually fixed something today with no human paged	A live counter, incremented every time a real rApp action fires across all six rApps in the Closed Loop simulation
Est. Annual OPEX Saved	Illustrative estimate of savings from energy management and reduced manual labor	A fixed illustrative figure — clearly not a live measurement, since a browser demo has no real cost data to draw from
Avg. Time to Detect & Resolve	How quickly, on average, an issue goes from appearing to being fixed	A fixed illustrative figure, framed against a "vs. hours, manual process" baseline for contrast

3.3 Operational KPI row

Active Alarms and Network Availability link through to the FCAPS Monitor tab; rApps in Closed Loop links through to the Closed Loop tab. All three are clickable, with a hover-highlight affordance, so the Overview page doubles as a navigation hub, not just a summary.

3.4 The cluster grid

Every function in the platform is grouped into eight clusters spanning its full functional scope. Each cluster card leads with a plain-English business-value line before the technical description, and is directly clickable through to its corresponding tab.

CLUSTER	LINKS TO TAB
RAN Architecture: RU/DU/CU & Interfaces	RAN Architecture
SMO Architecture & Interfaces	Data & Openness
FCAPS & Assurance	FCAPS Monitor
Cloud & Automation	Cloud & NF Lifecycle
Non-RT RIC: rApps, AI/ML, Agentic	Closed Loop
Near-RT RIC: xApps & Real-Time Control	Near-RT RIC
Slicing, Conflict & Intent	Slice Lifecycle
Security, Integration & Support	Security & Trust

4. RAN Architecture Tab

This tab exists to answer, visually, one of the most commonly asked architecture questions in this space: "How do SMO, the Non-RT RIC, and the Near-RT RIC divide responsibility, and what actually connects them?"

4.1 The diagram

Three layers, stacked top to bottom: SMO (hosting the Non-RT RIC / rApps) at the top, connecting to everything below via O1 (FCAPS), O2 (to cloud infrastructure), and A1 (policy, downward only); the Near-RT RIC in the middle, hosting xApps, connecting to the radio elements via E2; and the radio chain itself at the bottom — RU, connected via Open Fronthaul to DU, connected via F1 to CU-CP/CU-UP (themselves connected via E1), connected via backhaul to the Core network.

4.2 The hover-highlight interaction

Every interface label in the diagram, and every row in the interface reference table beneath it, is wired to a shared highlight function. Hovering an interface anywhere — in the diagram or the table — highlights its counterpart in both places simultaneously.

This interaction went through two rounds of correction. The first fixed a case where only part of the combined O1/O2/A1 line highlighted correctly. A second, later fix addressed a subtler bug: hovering an interface in the *diagram* correctly highlighted the matching row in the table, but moving the mouse away only cleared the diagram's own highlight, not the table row's — so the table row could stay highlighted indefinitely. The highlight-clearing logic was centralized so that every hover-out, from either the diagram or the table, resets every highlight everywhere, regardless of which element triggered it.

Every box in the diagram (RU, DU, CU-CP/CU-UP, Core, SMO, Near-RT RIC) also carries its own hover tooltip with a plain-English explanation, independent of the interface highlighting.

4.3 Interface reference table

See Section 2.3 above for the full, identical table used in this tab.

5. FCAPS Monitor Tab

This tab covers core network monitoring discipline, the day-to-day foundation every SMO is fundamentally built on. All four scrollable panels on this tab — Active Alarms, Live KPIs, Configuration Management, and Trace Sessions — share one consistent box size, so the tab's overall layout stays visually stable regardless of how much data has accumulated in any one panel.

5.1 Fault Management — active alarms

A scrollable list of 100 alarms, each with a severity level, a source (which cell or network function raised it), a description, a time-since-raised counter, and two independent actions: **Acknowledge** and **Cancel**.

SEVERITY	MEANING
Critical	Something has completely stopped working — e.g. a site outage
Major	Significant degradation that needs attention soon — e.g. sustained high load
Minor	A condition worth tracking but not yet service-impacting — e.g. rising CPU use
Warning	An early trend worth watching — e.g. a slowly rising failure rate

Acknowledging an alarm marks it as reviewed (it fades slightly and shows an "acknowledged" pill) without deleting it — reflecting how real fault management preserves a record even after a human has taken ownership of an issue. **Cancel** is a separate, distinct action that removes the alarm from the list entirely, once it's actually been resolved — the two actions are deliberately kept independent so acknowledging something doesn't imply it's gone, and clearing it doesn't require pretending it was never reviewed.

5.2 Performance Management — live KPIs

Ten KPIs tick continuously, each on a bounded random walk (see Section 5.2.1) so they feel alive without ever drifting to an unrealistic value. Hovering any KPI reveals a synthetic 24-hour trend line plus that trend's minimum and maximum points, generated in the KPI's real unit range and clearly labeled as illustrative — the live value shown on the card itself is the real-time one.

KPI	BOUNDS	WHY IT MATTERS
RAN Availability	99.5% - 100%	Availability is a percentage of uptime and can never exceed 100% — an earlier version of the demo had a bug where an unbounded random walk could drift the number above 100%, since fixed.
Call Setup Success Rate	95% - 100%	Reflects a healthy but not impossibly perfect network.
Avg. DL Throughput / cell	120 - 260 Mbps	A realistic range for a modern 5G cell under mixed load.
Drop Call Rate	0.05% - 2.5%	Kept low, since a healthy network should rarely drop calls, but never exactly zero.
Avg. UL Throughput / cell	15 - 60 Mbps	Uplink is typically a fraction of downlink capacity on the same cell.
PRB Utilization (network avg)	40% - 95%	How busy the radio capacity is across the network on average.
Handover Success Rate	90% - 100%	How cleanly devices move between cells as they travel.
RRC Setup Success Rate	95% - 100%	How reliably a device establishes its initial radio connection, distinct from a later handover.
Packet Error Rate (URLLC slice)	0.001% - 0.5%	The near-zero-tolerance metric for the Ultra-Reliable Low-Latency slice specifically.
Avg. Latency (PDB compliance)	2 - 15 ms	Measured against the Packet Delay Budget — the max acceptable one-way delay before it's a QoS violation.

5.2.1 Why a "bounded random walk"?

Rather than picking a new random number every tick (which would look jumpy and unrealistic), each KPI takes a small random step from its current value, then that step is clamped to the KPI's realistic min/max range. This produces a value that drifts smoothly, the way a real network's performance actually does, while never producing an impossible reading. The same underlying real-unit range is reused to generate each KPI's synthetic 24-hour hover trend, so the trend and the live value are always on a consistent scale.

5.3 Configuration Management — recent changes

A scrollable log of 100 configuration-change entries, each showing the timestamp, which network entity was affected, which parameter changed, the old and new values, and who or what made the change. Several entries are attributed to specific RApps

(e.g. "Energy Saving Management" changing a power parameter), directly illustrating that automated systems, not only human engineers, are configuration change sources that need to be tracked and auditable.

5.4 Trace sessions

A "trace" follows one specific call or device through the network in detail, used for deep troubleshooting when a specific customer's experience needs to be understood precisely. The tab allows starting a new trace against a randomly selected cell, which transitions from "running" to "complete" after a short delay. Hovering a completed session reveals a synthetic trace analysis snapshot — RRC setup time, DL/UL throughput, packet error rate, handover count, and session duration — clearly labeled as an illustrative snapshot rather than a decoded real trace file. The trace table shares the same fixed box size as the other three panels on this tab, so it scrolls rather than growing the page as more sessions are started.

5.5 Topology & inventory summary

Three rollup counts: number of sites, number of radio network functions (RU/DU/CU instances), and number of distinct RAN vendors integrated — representing the "single source of truth" inventory that every other automation function in the platform ultimately depends on being accurate.

6. Closed Loop Tab & Scenario Reasoning

This is the centerpiece of the demo: a live visualization of the Non-RT RIC actually detecting problems and fixing them automatically, with real O-RAN SON-style algorithms behind each fix rather than placeholder logic — now across all six rApps, not four.

6.1 The cell ring

Twelve cells, drawn from three different simulated RAN vendors, arranged in a ring around a central "Non-RT RIC" hub. Each cell is colored by its current state (green = nominal, amber = congested, red = alarm, cyan = an rApp is actively remediating it), with an additional colored outer ring indicating vendor. Hovering any cell shows its ID, vendor, state, live PRB/handover metrics, and the last rApp action taken on it. The cyan "actively remediating" indicator now persists for a few seconds after any action fires, rather than clearing within the same simulation tick it appeared in — an earlier version resolved some fixes (like Cell Outage Compensation) so quickly that the visual confirmation was, in practice, too brief to actually see.

6.2 The two view modes: sample vs. fleet scale

A toggle switches between the 12-cell interactive ring and a 10,000-cell "fleet view," using a canvas-rendered density heatmap with rollout statistics and a ranked exceptions list rather than 10,000 individual dots — the correct pattern at scale is aggregation with drill-down, not enumeration.

6.3 The six trigger scenarios — detailed reasoning

Each button injects a specific, realistic fault or performance condition onto a randomly chosen cell, color-coded and matched one-to-one to the rApp that resolves it. Clicking a button produces three layers of confirmation feedback: the button flashes, a toast notification describes exactly what happened and to which cell, and a shockwave ring pulses outward from the affected cell in the diagram.

Simulate load spike (MLB)

What it does: Sets a randomly chosen cell's PRB utilization to 88% and marks its state as "congested."

Real-world meaning: A sudden surge in demand on one tower — a large public event, a stadium emptying out, a viral moment.

What responds: Mobility Load Balancing detects PRB utilization above its 75% threshold and offloads a portion of idle-mode users to neighboring, less-busy cells.

Simulate sleeping cell (ESM)

What it does: Sets a randomly chosen cell's PRB utilization to 8%, keeping its state "nominal."

Real-world meaning: A cell sitting mostly idle — a business district at 3am, a light-traffic rural site overnight.

What responds: Energy Saving Management detects sustained low utilization below its 20% threshold and applies a micro cell-sleep mode. This remains the single most quantifiable, provable value case for rApp automation in the whole industry.

Simulate cell outage (COC)

What it does: Sets a randomly chosen cell's state to "alarm" and its PRB utilization to 95%.

Real-world meaning: A hard site failure — power outage, hardware fault, fiber cut.

What responds: Cell Outage Compensation adjusts neighboring cells' antenna tilt to widen coverage and partially fill the gap, while the fault is separately raised in Fault Management for a human to physically repair.

Simulate handover failures (MRO)

What it does: Pushes a randomly chosen cell's handover failure rate to 5.5%.

Real-world meaning: A cell whose neighbor relations or radio parameters have drifted out of tune, so devices increasingly fail to hand over cleanly as they move between towers.

What responds: Mobility Robustness Optimization detects the rate above its 3.0% threshold and incrementally increases handover hysteresis until the rate settles back down, typically over a few simulation cycles.

Simulate missing neighbor (ANR)

What it does: Flags a randomly chosen cell as missing a neighbor relation across an EMS boundary.

Real-world meaning: Two cells, often from different vendors' management systems, that should know about each other for handover purposes but don't yet — normally only found when a customer complains about a dropped call at a specific location.

What responds: Automatic Neighbor Relation discovers and adds the missing relation.

Simulate PCI collision (PCI)

What it does: Flags a randomly chosen cell with a Physical Cell Identity collision.

Real-world meaning: Two nearby cells accidentally broadcasting the same radio ID, confusing devices and causing intermittent connection problems that are notoriously hard to diagnose from symptoms alone.

What responds: PCI Reuse Optimization detects the collision, reassigns an ID respecting the minimum reuse distance, and re-verifies PRACH consistency afterward.

Reset to nominal

Regenerates all 12 cells back to random, healthy baseline values, clearing any injected condition — a demo-usability feature styled as a neutral action since it isn't injecting any condition.

6.4 Live conflict detection

The six trigger conditions are designed to be independent, but they aren't mutually exclusive by construction. Every simulation cycle, the platform now checks whether more than one enabled rApp genuinely wants to act on the same cell at the same time — natural random drift, or triggering more than one scenario in quick succession, can occasionally cause exactly that. When it happens, it's a real detected conflict, not a scripted one: it's logged live to the Conflict Log tab and resolved using whichever of the three strategies (Allow / First-Come-First-Served / Priority-Based Override) is currently selected there. See Section 11 for the full resolution logic and the priority ordering used.

6.5 Copilot narration (plain-language vs. technical log toggle)

The R1 interface event feed can be switched between a raw technical log and a plain-language narration of the same event — demonstrating that agentic and autonomous systems are expected to explain their own decisions in a form a non-specialist can audit, not operate as an unexplained black box.

7. Near-RT RIC Tab

This tab exists specifically to close a gap: without it, the whole demo would only ever show the slow half of RAN intelligence (the Non-RT RIC), when a complete picture requires showing both halves and, critically, the actual timescale difference between them.

7.1 The explicit contrast

Everything on the Closed Loop tab is the Non-RT RIC, running on a greater-than-1-second (often minutes+) timescale. This tab is the Near-RT RIC, running on a 10-millisecond-to-1-second timescale over the E2 interface. The Non-RT RIC still steers the Near-RT RIC, but only via slower A1 policy guidance, not by micromanaging every millisecond itself.

7.2 Making the timescale difference visible, not just stated

A live "E2 Control Actions / sec" counter ticks far faster than the rApp action counter on the Closed Loop tab — one E2 control tick every 150 milliseconds, versus the Non-RT RIC's roughly one action every few seconds — making the distinction impossible to miss.

7.3 The xApp catalog

XAPP	WHAT IT DOES
Dynamic Spectrum Sharing	Real-time arbitration of shared spectrum between 4G and 5G carriers, slot by slot
Real-Time Load Balancing	Millisecond-scale traffic steering between cells based on instantaneous radio conditions
Interference Mitigation	Detects and reacts to interference within the same control interval it is observed in
QoS-Aware Scheduling Assist	Adjusts scheduler weighting in real time to protect latency-sensitive traffic flows

8. rApp Catalog Tab

All six rApps modeled in this demo now have genuine, runnable reference logic — real O-RAN SON-style automation, grounded in generic industry requirement categories rather than any single vendor's proprietary product. This closes a gap from the previous edition: Automatic Neighbor Relation and PCI Reuse Optimization previously had no working trigger condition anywhere in the simulation, so enabling them had no observable effect; both now respond to a dedicated condition (see Section 6.3) exactly like the other four.

RAPP	CATEGORY	WHAT IT ACTUALLY DOES (LIVE LOGIC)
Energy Saving Management (ESM)	AI/ML forecast	If a cell's PRB utilization stays below 20% for a sustained period, applies a micro cell-sleep mode, reducing power draw.
Automatic Neighbor Relation (ANR)	Configuration Management	Discovers and adds a missing neighbor relation across an EMS boundary whenever one is flagged on a cell.
Mobility Robustness Optimization (MRO)	Performance Management	If a cell's handover failure rate exceeds 3.0%, increases handover hysteresis by 1dB, smoothing out the transition and reducing failures.
Mobility Load Balancing (MLB)	Performance Management	If a cell's PRB utilization exceeds 75%, offloads roughly 10% of idle-mode users to a neighboring, less-busy cell.
Cell Outage Compensation (COC)	Fault Management / Self-Healing	If a cell enters alarm state, adjusts a neighboring cell's antenna tilt to widen coverage and partially fill the resulting gap.
PCI Reuse Optimization (PCI)	Configuration Management	Detects a flagged Physical Cell Identity collision between nearby towers and reassigns within the minimum reuse distance.

All six rApps are enabled by default. Each card also shows which R1-style data services it subscribes to (e.g. cell metrics, PRB utilization) — reflecting the real O-RAN principle that an rApp declares its data dependencies explicitly through the R1 interface (via SME/DME) rather than reaching into arbitrary network data directly.

9. Import rApp Tab

This tab demonstrates the "bring your own rApp" capability that is central to O-RAN's open-ecosystem vision.

9.1 The descriptor schema

Modeled on real Non-RT RIC rApp Manager conventions, a valid descriptor must declare: `rAppId`, `name`, `version`, `requiredServices`, `exposedServices`, `policyType`, and a `logic` object naming an algorithm and its parameters.

9.2 Honest handling of unrecognized logic

If the submitted descriptor's `logic.algorithm` matches one of the six algorithms the demo actually implements — energy saving, mobility robustness, load balancing, outage compensation, neighbor discovery, or PCI reuse — the imported rApp runs genuine logic identical to the built-in versions. If it names anything else, the rApp is honestly onboarded as a monitoring-only stub rather than silently pretending to execute logic that doesn't exist.

9.3 Why this reflects a real technical nuance

In an actual O-RAN deployment, a package descriptor mostly describes deployment metadata, while the actual decision algorithm lives in the rApp's own proprietary code. The demo's descriptor format mirrors that real convention, while adding a logic block purely so the demo can actually execute something when a recognized algorithm name is submitted.

10. Slice Lifecycle Tab

A slice is not a one-time configuration — it is a living, managed entity across its whole life. This tab demonstrates that full lifecycle rather than only a static "here is a slice" snapshot.

10.1 The four lifecycle stages

STAGE	WHAT HAPPENS
1. Create	NSSI creation request accepted, S-NSSAI allocated, RAN resource template applied
2. Modify	NSSI modification applied without service disruption — e.g. a PRB partition updated at runtime
3. Monitor & Assure	Continuous SLA monitoring against the slice's committed targets
4. Decommission	NSSI decommission executed, resources released, inventory reconciled

10.2 Where each stage's inputs actually come from

New in this edition: clicking through the four stages now updates a dedicated panel documenting exactly what input each stage needs and where it would come from in a real deployment, rather than treating "lifecycle" as an unexplained black box.

STAGE	INPUT SOURCE, AS DOCUMENTED IN THE DEMO
Create	Either the "Express a business intent" panel on this same tab — an operator picks a plain-language intent, translated into S-NSSAI / 5QI / PRB policy — or, in a real deployment, a formal slice order from the NSMF/OSS layer.
Modify	A manual operator change request, or an automated closed-loop recommendation from a Non-RT RIC rApp reacting to SLA drift detected during Monitor & Assure.
Monitor & Assure	The same Performance Management counter stream shown on FCAPS Monitor, filtered down to this slice's S-NSSAI — no separate telemetry path.
Decommission	An OSS/BSS service-termination order, or an automatic expiry (e.g. a temporary "event-mode" slice tearing itself down once its event window closes).

10.3 Business intent translation

Three example business intents (guarantee low latency for an enterprise customer, guarantee sustained throughput for a premium consumer tier, maximize capacity for a stadium event) each translate into concrete technical policy — a specific 5QI value, a target Packet Delay Budget, and a scheduling priority weight — with a live bar chart showing actual performance against the committed SLA target for each active slice.

11. Conflict Log Tab

When multiple rApps act on the same network simultaneously, their recommendations can directly contradict each other. This tab demonstrates detecting and resolving exactly that kind of conflict.

11.1 Two sources of conflicts, not one

A manual "simulate a conflict now" button demonstrates a scripted, same-parameter conflict between two named rApps (Energy Saving Management and Mobility Load Balancing). New in this edition: the Closed Loop tab also detects real conflicts live, checking every simulation cycle whether more than one enabled rApp genuinely wants to act on the same cell at the same time, and logging it here automatically when it happens — using whichever strategy is currently selected below, the same as the manual button.

11.2 The three resolution strategies

STRATEGY	BEHAVIOR
Allow	Both (or all) conflicting configurations are applied; the SMO simply monitors the resulting KPI drift rather than blocking any action.
First-Come-First-Served (FCFS)	Whichever rApp's action was applied first is retained; the others are rejected for a cooldown window.
Priority-Based Override	A pre-configured priority ranking decides which rApp's action wins outright, regardless of timing.

11.3 Priority order used for real, detected conflicts

When Priority-Based Override resolves a genuinely-detected conflict from the Closed Loop tab, the ranking (highest priority first) is: Cell Outage Compensation, Mobility Robustness Optimization, Mobility Load Balancing, Energy Saving Management, Automatic Neighbor Relation, PCI Reuse Optimization — reflecting roughly how severe each underlying problem is in a real network.

11.4 Why this matters

Without conflict management, multiple well-intentioned automated systems can fight each other indefinitely. This is widely regarded as one of the areas of O-RAN standards still actively maturing, rather than a fully solved problem industry-wide.

12. Data & Openness Tab

12.1 Data coverage matrix

A grid showing, per data domain (Fault Management, raw Performance counters, Configuration Management, Topology & Inventory) and per simulated vendor, whether that data is exposed through an open interface or retained internally by that vendor only.

12.2 Resilience / decoupling simulator

A button simulates the external data platform becoming completely unreachable. A well-decoupled SMO architecture should continue buffering events locally and keep its closed loops stable through the outage. This directly feeds the "Decoupled architecture" score on the Openness Scorecard tab (see Section 16).

12.3 Self-serve data routing

A simple form lets a user redirect a data domain to a different destination endpoint, applied immediately with no simulated ticket or maintenance window — illustrating the difference between a genuinely open platform and a closed one.

13. Cloud & NF Lifecycle Tab

This tab was substantially reworked in this edition, moving from five generic, always-available action buttons to a genuinely interactive, select-an-object model.

13.1 Infrastructure visibility table

A live, scrollable inventory of 100 infrastructure objects — vDU clusters, vCU-CP/vCU-UP instances, Near-RT RIC hosts, and O-Cloud worker nodes — spread across seven simulated locations (edge sites, regional data centers, and central cloud regions). Clicking any row selects it, shown with a highlighted background and an indicator in a "selected object" banner above the action buttons.

13.2 Network function lifecycle actions

Five actions — Instantiate, Heal, Scale, Upgrade, Terminate — but now with real conditions attached rather than always simply "completing":

ACTION	BEHAVIOR
Instantiate	Always available; creates a brand-new object (not tied to the current selection) and automatically selects it once it finishes standing up.
Heal	Requires a selected object. Only does something if that object's status is actually "degraded" — otherwise it reports there was nothing to fix.
Scale / Upgrade	Require a selected object; transition it through a visible "scaling"/"upgrading" state before returning to "running."
Terminate	Requires a selected object; transitions through "terminating" before removing it from the inventory.

13.3 Live, self-drifting status

Status is no longer static between clicks. Periodically, a running object can spontaneously degrade on its own — the way real infrastructure actually behaves — giving a genuine reason to use Heal rather than Heal always just being decorative. Separately, any transient state (scaling, upgrading, healing, instantiating) automatically settles back to "running" after roughly nine seconds even if left untouched, so nothing stays stuck mid-transition indefinitely, while a "degraded" object stays degraded until deliberately healed.

13.4 Where an Instantiate request's data actually comes from

A new explanatory note on this tab documents the real-world provenance of a new function: the software package and its signature come from the operator's container/artifact registry; the site's physical hardware is auto-discovered over O1 Plug-and-Play once powered on-site, rather than typed in by hand; the configuration is generated from planning-tool design inputs via one common templating engine; and the post-launch KPI validation window reads live PM counters from the same feed shown on FCAPS Monitor. In practice, an Instantiate request originates from a Zero-Touch Rollout pipeline run, a Slice Lifecycle capacity change, or a manual trigger on this tab — and whichever it is, the new object appears immediately, live, in the infrastructure table described in 13.1.

14. Zero-Touch Rollout Tab

This tab models a full, thirteen-step commissioning pipeline for bringing a new site online with minimal manual intervention.

14.1 The thirteen steps

1. Onboard software package to the SMO — validate format, signature, and contents against the container registry
2. Auto-discover on-site hardware (RU/DU/CU) via O1 plug-and-play
3. Synchronize the new cloud instance / network element with the SMO inventory
4. Generate site configuration from design inputs (a single engine, all architectures)
5. Create a CNF deployment plan and map configuration values from the generated site config
6. Deploy the CNF to the O-Cloud via O2-DMS, with auto-configuration if PnP is supported
7. Load the site configuration to the network element for commissioning
8. Validate the configuration plan against schema before activation
9. Activate the configuration plan
10. Verify software version and inter-node (e.g. CU-DU) connectivity
11. Pre-launch RF/coverage check
12. Post-launch KPI validation window (holds if any metric is out of bounds)
13. Save the activated configuration as the new baseline; generate a step-by-step audit report

14.2 Where a new integration's data comes from

New in this edition: an explanatory note makes the data provenance for each phase explicit, rather than leaving "zero-touch" implying the data appears from nowhere — see Section 13.4, which documents the identical chain shared with Cloud & NF Lifecycle.

14.3 Why this level of granularity matters

Each step is individually animated and logged, specifically to make the point that "zero-touch" does not mean "no process" — it means a rigorous, auditable, automatically-executed process, with validation gates and rollback points built in.

15. Security & Trust Tab

15.1 Evidence-based security posture

CONTROL	EVIDENCE SHOWN WHEN ENABLED
Zero-trust access (NIST SP 800-207)	Every service call authenticated and authorized — zero implicit-trust paths found
Component hardening (CIS Benchmarks)	Database/web-server images scanned against a specific CIS baseline version, with findings remediated
Supply-chain integrity	SBOM generated, image signatures verified, zero unverified artifacts in the deployment path
Secrets management	Encrypted at rest and in transit, access logged, credentials rotated on a fixed schedule

The deliberate design principle here is "evidence, not assertion" — a mature platform should be able to show proof of a control working, not merely claim the control exists.

15.2 Supply-chain artifact verification

A button verifies the next artifact in a rotating list (an rApp package, a Helm chart, a container image, an SBOM manifest), logging a signature/checksum verification against its build-time attestation each time.

New in this edition: the tab now explicitly explains what the check actually does (a cryptographic signature and checksum comparison, proving the artifact wasn't tampered with) and, importantly, *how the next artifact is chosen* — not randomly, but the next item in a transparent, round-robin deployment queue. A live "next in queue" indicator shows exactly which artifact will be checked next, so the selection mechanism isn't a mystery.

16. Openness Scorecard Tab

Six industry-relevant principles, each scored 0-100, rolling up into a single composite score. There is no live sensor for "how open is this architecture" — these are inherently qualitative judgment calls. Rather than inventing a fake precise formula, each principle is wired to something genuinely measurable elsewhere in the demo.

PRINCIPLE	LIVE DATA SOURCE, AND WHAT ACTION MOVES IT
Openness & data ownership	Percentage of the Data & Openness coverage matrix marked "exposed" — toggle rows there to change it.
Decoupled architecture	Rises with each successful reconnect after a simulated data-platform outage on the Data & Openness tab.
Trustworthy automation	Percentage of currently active rApps running genuine (non-stub) reference logic, versus imported-but-unrecognized stubs — enable/disable rApps on Closed Loop or the Catalog to change it.
Security by evidence	Percentage of Security & Trust controls currently showing evidence — toggle controls on that tab.
Safe autonomy (conflict-aware)	Strength of the currently active conflict-resolution strategy on the Conflict Log tab.
Freedom from lock-in	Percentage of the Overview's functional clusters that are actually interactive versus documentation-only.

16.1 What happens when you drag a slider

Dragging a slider overrides that one dimension with a manual judgment call — useful for scoring a real vendor rather than this demo — and immediately recalculates the composite score below from that override. It does not change anything else in the demo. That manual value stays in place until either dragged again or the "Recalculate from live demo state" button is pressed, which discards every manual override and pulls all six dimensions back from whatever is currently true elsewhere in the demo.

17. Documentation Tab

New in this edition. This tab is the in-app counterpart to this document — a full written reference living inside the demo itself, so a reader never has to leave the browser tab to understand what they're looking at.

17.1 Structure

The tab is organized into five jump-to sections: a Glossary (matching Section 2 of this document, including the 5G traffic-type and cloud/security subsections), a Tab-by-tab guide (a condensed version of Sections 3–16), Scenario reasoning (matching Section 6.3–6.4, now covering all six trigger buttons plus the live conflict-detection mechanic), Design principles (matching Section 19), and a two-column Acronym index.

17.2 Acronym index ordering

The acronym index lists all 42 acronyms used anywhere in the demo, sorted alphabetically as a single continuous sequence split across two columns — reading down the left column continues seamlessly at the top of the right column, rather than alternating left-right within each row. See Section 20.2 for the identical, full alphabetized list.

18. 50 Design Questions Tab

New in this edition. A public-standards-based series of 50 design-evaluation topics for assessing any SMO platform — not just this one — grounded in O-RAN Alliance, 3GPP, TM Forum, ETSI, NIST, and CIS material rather than any vendor's internal documentation.

18.1 Structure

The 50 topics contain 224 specific sub-questions in total, grouped into 8 themes: Openness & Data Ownership; Decoupled Architecture (Cloud & Infrastructure); Trustworthy Automation (Zero-Touch); Trustworthy Automation (Self-Healing & Optimization); Safe Autonomy (Network Slicing); Safe Autonomy (Application Ecosystem); Safe Autonomy (Conflict Management); and Security by Evidence & Freedom from Lock-In. Each topic links directly to the tab where this demo answers it, and a keyword filter narrows the list live.

18.2 Honesty about gaps

Several topics are answered "partially" (e.g. RAN-side slice lifecycle is shown, but independent deployability from core/transport slicing is not) and one — RAN sharing independence across multiple operators — is marked as genuinely out of scope for a single-operator browser demo, rather than every topic being forced into a claimed full answer.

18.3 What changed with this edition

Several answers were revisited and expanded to reflect the interactivity added since the previous edition — most notably the questions covering NF lifecycle automation and infrastructure visibility (now describing the select-an-object model and self-drifting status), self-healing (now describing the visible cyan pulse fix and the parallel infrastructure self-healing example), slice lifecycle (now crediting the per-stage input-provenance panel), and conflict detection/resolution (now describing the live, automatic detection mechanism alongside the original manual demo button).

19. Design Principles

19.1 One file, no backend

Every tab, every simulation, every calculation runs client-side in a single HTML file with embedded CSS and JavaScript. Nothing can fail due to network issues, API rate limits, or a backend going down mid-conversation.

19.2 Honesty about synthetic data

Every screen in the demo is explicitly labeled as simulation. The Import rApp tab labels unrecognized imported logic as a "monitoring-only stub" instead of silently pretending to run behavior that doesn't exist, and the top-level badge on every page states "SIMULATION — SYNTHETIC DATA" without qualification.

19.3 Grounded in public standards only

All terminology, interface names, and behaviors are drawn from public O-RAN Alliance, 3GPP, TM Forum, NIST, and CIS material. No proprietary vendor documentation or trademarked commercial product names are used anywhere in this demo.

19.4 Scale reasoning: aggregation over enumeration

Individual per-object visualizations become both computationally impractical and cognitively unreadable well before real network scale. The correct pattern is aggregation with drill-down — a density heatmap, rollout statistics, and a ranked exceptions worklist — with the detailed per-cell ring view reserved for whatever a user actually needs to drill into.

19.5 Executive readability alongside technical depth

Plain-language framing throughout — an "In plain terms" callout on the Overview page, business-value one-liners preceding every technical description, and a comprehensive hover-tooltip system covering every tab name, KPI title, and jargon abbreviation — without removing or diluting the technical depth available to someone who wants it.

19.6 Visible confirmation of every action

Every user-triggered action produces layered, immediate feedback — a button-level animation, a toast notification with the specific outcome, and (for the Closed Loop scenarios) a shockwave highlight directly on the affected element.

19.7 Real detection, not scripted outcomes (new)

Where the demo shows a conflict or a fault, it's because the underlying simulation state actually produced one — e.g. the Closed Loop tab checks every cycle whether two enabled rApps genuinely want to act on the same cell at the same time, rather than playing back a canned example.

19.8 State that resolves on its own, imperfectly (new)

Infrastructure objects can spontaneously degrade and transient states settle back to normal after a while even without intervention — closer to how real infrastructure actually behaves than a demo where nothing changes unless a user clicks something.

19.9 A fixed, predictable layout footprint (new)

Scrollable panels are sized consistently so the page doesn't visibly grow or shift as more alarms, sessions, or configuration entries accumulate during a live walkthrough.

20. Appendix: Full Interface, rApp & Acronym Reference Tables

20.1 Interface quick reference

INTERFACE	CONNECTS	TIMESCALE / NOTES
O-FH	RU ↔ DU	Fronthaul, microsecond-scale, functional split 7.2x
F1	DU ↔ CU	Midhaul, single-digit ms
E1	CU-CP ↔ CU-UP	Internal CU split
E2	Near-RT RIC ↔ RAN	10ms-1s, xApp control channel
A1	Non-RT RIC ↔ Near-RT RIC	Policy guidance, greater than 1s
O1	SMO ↔ RAN elements	FCAPS
O2	SMO ↔ O-Cloud	Infrastructure/workload lifecycle
R1	rApps ↔ Non-RT RIC	Data/service exposure (SME/DME) — still maturing in O-RAN standards

20.2 rApp quick reference (all 6, alphabetical)

RAPP	TRIGGER CONDITION IN THIS DEMO
Automatic Neighbor Relation (ANR)	A cell flagged as missing a neighbor relation
Cell Outage Compensation (COC)	A cell in "alarm" state
Energy Saving Management (ESM)	Sustained PRB utilization below 20%
Mobility Load Balancing (MLB)	PRB utilization above 75%
Mobility Robustness Optimization (MRO)	Handover failure rate above 3.0%
PCI Reuse Optimization (PCI)	A cell flagged with a PCI collision

20.3 Full acronym index (alphabetical)

ACRONYM	FULL TERM	ACRONYM	FULL TERM
5QI	5G QoS Identifier	O-FH	Open Fronthaul
ANR	Automatic Neighbor Relation	OPEX	Operating Expenses
BSS	Business Support System	OSS	Operations Support System
CA	Carrier Aggregation	PCI	Physical Cell Identity
CIS	Center for Internet Security	PDB	Packet Delay Budget
CNF	Cloud-native Network Function	PER	Packet Error Rate
COC	Cell Outage Compensation	PnP	Plug-and-Play
DL / UL	Downlink / Uplink	PRB	Physical Resource Block
DME	Data Management & Exposure	RAN	Radio Access Network
eMBB	Enhanced Mobile Broadband	RF / PHY	Radio Frequency / Physical Layer
ESM	Energy Saving Management	RIC	RAN Intelligent Controller
FCAPS	Fault, Config, Accounting, Performance, Security	RU/DU/CU	Radio/Distributed/Centralized Unit
GBR	Guaranteed Bit Rate	SBOM	Software Bill of Materials
HO	Handover	SLA	Service Level Agreement
MLB	Mobility Load Balancing	SME	Service Management & Exposure
mMTC	Massive Machine-Type Communication	SMO	Service Management & Orchestration
MRO	Mobility Robustness Optimization	S-NSSAI	Slice Selection Assistance Info
NF / LCM	Network Function / Lifecycle	SON	Self-Organizing Network

Mgmt

NIST	National Institute of Standards & Technology	TCO	Total Cost of Ownership
NSSI	Network Slice Subnet Instance	UE	User Equipment
O2 - DMS	O2 Deployment Management Service	URLLC	Ultra-Reliable Low-Latency Communication